Scalable fit of the output-error sensitivity of a Clements unitary mesh vs. phase-shifter error

large range $[0, 0.5]$ rad with the small-range parameters (no refit)

Model (overlap-compatible, scalable in N/N_0): $E_\phi(\sigma_{\mathrm{rad}}, N) = \sqrt{2}\,\sqrt{1 - e^{-u_\phi^2}}$, $u_\phi = \dfrac{s_\phi(N)\,\sigma_{\mathrm{rad}}}{\sqrt{2}}$, $s_\phi(N) = s_{\phi,0}\,(N/N_0)^{p_\phi}$, $N_0 = 100$

$s_{\phi,0}$ = initial slope $dE/d\sigma_{\mathrm{rad}}$ at the reference size $N_0 = 100$ (rad^{-1})

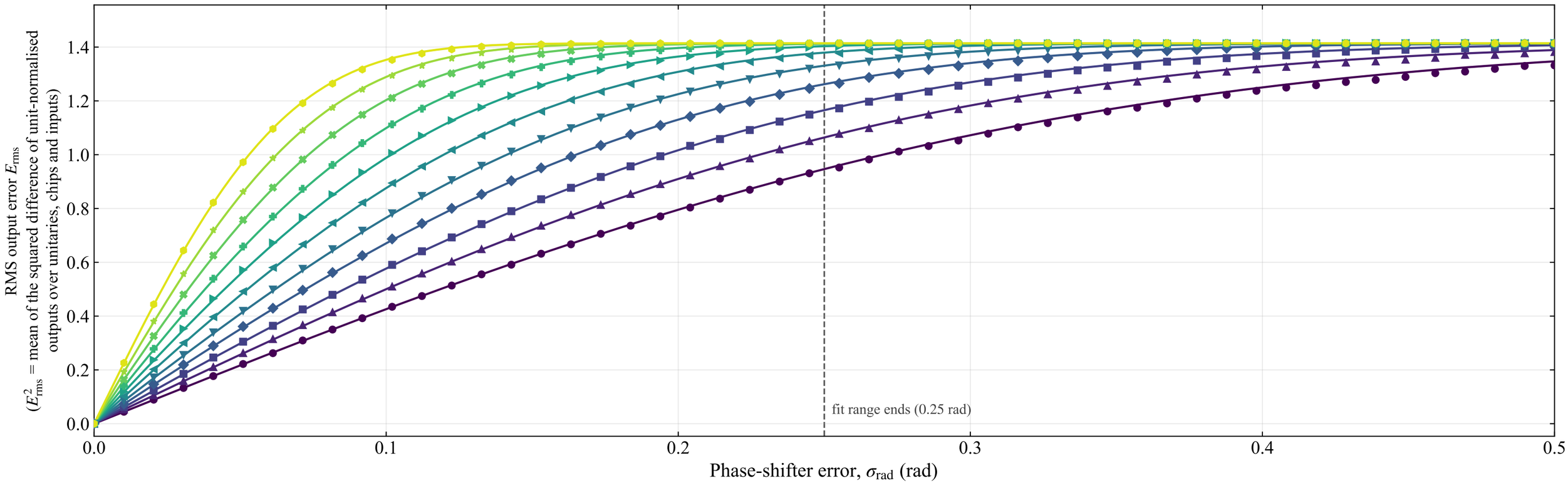
p_ϕ = network-size scaling exponent of the sensitivity ($p_\phi \approx 0.5$: $\sqrt{N}$ growth)

Best-fit values (small range, all N, equal weight per point; $\pm$ = bootstrap s.d. over unitaries): $s_{\phi,0} = 9.83706 \pm 0.00245$ rad^{-1}, $p_\phi = 0.50554 \pm 0.00035$

Quality on this range (parameters NOT refitted): all 550 points: RMSE = 0.00436, MAE = 0.00301, $R^2 = 0.99989$, max$|\Delta E| = 0.0180$ | off-plateau ($E \le 1.2$, 206 points): RMSE = 0.00345, $R^2 = 0.99991$, mean rel. err. = 0.46%

Hardware model: i.i.d. Gaussian phase errors $\delta\theta, \delta\phi \sim \mathcal{N}(0, \sigma_{\mathrm{rad}}^2)$ on every MZI; all other channels ideal | markers = simulated E_{rms} (over unitaries, chips and input types), solid lines = scalable fit (fit in E space)

$N = \{20, 28, 38, 53, 72, 100, 138, 190, 263, 362, 500\}$ | Haar-random unitary U (single Clements mesh) | 7 input types × 10 draws, pooled (equal weight) | one forward pass $y = Ux$ | $E_{\mathrm{rms}} = \sqrt{\langle \|\hat{y}^{\mathrm{hw}} - \hat{y}^{\mathrm{ideal}}\|_2^2 \rangle}$ | 25 unitaries × 30 chips | 50 points per curve



Network size N | scaled sensitivity from the fit (markers: data, lines: fit)

$N = 20$, $s_\phi = 4.360$ rad^{-1}	$N = 53$, $s_\phi = 7.136$ rad^{-1}	$N = 138$, $s_\phi = 11.577$ rad^{-1}	$N = 362$, $s_\phi = 18.850$ rad^{-1}
$N = 28$, $s_\phi = 5.169$ rad^{-1}	$N = 72$, $s_\phi = 8.332$ rad^{-1}	$N = 190$, $s_\phi = 13.608$ rad^{-1}	$N = 500$, $s_\phi = 22.193$ rad^{-1}
$N = 38$, $s_\phi = 6.032$ rad^{-1}	$N = 100$, $s_\phi = 9.837$ rad^{-1}	$N = 263$, $s_\phi = 16.039$ rad^{-1}	